

Data Center & Digital Infrastructure: Reading the AI Capex Cycle from the Wafer Inward

AN INSTITUTIONAL REAL ESTATE INVESTMENT THESIS

Draft — independent research. Figures current as of Q1 2026 earnings; verify against latest filings before circulation.

Executive Summary

The consensus framing of the AI infrastructure trade starts at the data center floor and reasons outward — hyperscaler capex announcements drive headlines, and data center REITs trade as a derivative of those announcements. This thesis argues the more durable read runs the other direction: the upstream semiconductor supply chain prices and commits capital twelve to eighteen months ahead of the demand that shows up in REIT leasing, and that lead time is the edge available to a real estate investor willing to track it.

The signal is currently unambiguous. TSMC raised 2026 capital expenditure guidance to \$52–56 billion, near the top of its range and roughly 32% above 2025, with 70–80% directed at the advanced nodes (7nm and below) that exclusively serve AI and high-performance computing — nodes that already represent 74% of its wafer revenue (TSMC Q1 2026). The four largest US hyperscalers have collectively guided 2026 capital expenditure toward \$700 billion, up approximately 75% from the roughly \$410 billion record set in 2025, with about three-quarters earmarked for AI infrastructure (Financial Times compilation of Q1 2026 earnings). When the foundry at the base of the supply chain commits a third more capital against multi-year customer reservations, it is corroborating that hyperscaler spending is contracted demand rather than guidance theater.

That demand is colliding with a supply base that cannot respond on the same timeline. Advanced-node wafer capacity (7nm and below) is projected to grow only about 69% through 2028 — per SEMI's 300mm Fab Outlook — against demand growing well ahead of it; powered-shell vacancy in Northern Virginia sat below 5% in early 2025 and roughly 1% in the tightest submarkets; and announced capacity is increasingly slipping on power rather than chips — roughly a third of expected 2025 capacity was delayed or deferred, with an estimated 30–50% of 2026 capacity at risk (Sightline Climate). AI-ready campuses now cost \$1–2 billion and take 18–36 months to deliver. Supply is structurally short and slow, which is the condition under which real estate cash flows compound.

The investable conclusion is that data center REITs offer the cleanest public-market exposure to the AI capex cycle without single-name chip risk, and that the supply constraint — power, not silicon — cushions the downside that a pure technology-demand thesis would carry. The body of this piece works the demand signal forward through five stages of the supply chain, models the translation into data center take-up, and assesses Digital Realty (DLR), Equinix (EQIX), and Iron Mountain (IRM) on a risk-adjusted basis. The recommendation is to own the data-center REIT complex as a basket — Equinix, Digital Realty, and Iron Mountain — sized as a 5–10% thematic AI-infrastructure sleeve and accumulated into the power-headline selloffs that periodically hit the group rather than chased after its ~40% trailing run. On a risk-adjusted basis I weight Equinix as the core holding (the stickiest moat, the cleanest investment-grade balance sheet), Digital Realty as the purest investment-grade expression of the hyperscale thesis, and Iron Mountain as a smaller, higher-beta position whose best-in-group trailing return carries the group's most balance-sheet risk. The edge is the lead documented below: the upstream supply chain commits capital twelve to eighteen months ahead of REIT demand, and that signal — TSMC's 32% 2026 capex step-up and a second doubling of advanced packaging — is currently pointing to a further demand leg into 2026–27.

Section 1 — The Macro Framework: Five-Stage Capex Propagation

The error most data center analysis makes is treating hyperscaler capital expenditure as the leading indicator. By the time Microsoft or Amazon raises capex guidance on an earnings call, the order has already propagated through three or four upstream layers of the semiconductor supply chain, each of which committed capital and signaled intent months earlier. The investor who tracks the chain from its base — the foundry — sees the demand inflection before it reaches the real estate layer, where it ultimately prices.

The chain has five stages, each leading the next by a measurable interval.

Stage 1 — Foundry capex (12–18 month lead)

Advanced-node foundry capacity is the ultimate constraint on AI compute, and TSMC is effectively the entire constraint at the leading edge. Building a fabrication facility takes three to five years and upwards of \$20 billion, so foundry capital commitments are the longest-dated and least reversible signal in the chain. When TSMC commits capital, it is doing so

against customer prepayments and long-term capacity reservations — the company is lead-time disciplined and does not build speculative capacity without an anchor customer.

The current signal is maximally bullish. TSMC's 2026 capex guidance of \$52-56 billion represents a roughly 32% increase over 2025, with management explicitly attributing the step-up to AI demand and directing 70-80% of the spend to advanced process nodes and another 10-20% to advanced packaging (TSMC Q1 2026 earnings). Critically, the company is doubling CoWoS packaging capacity — the specific bottleneck for high-performance AI accelerators — and adding N3-capable fabs against a stated policy of not normally expanding a node past its target capacity. The metric to track quarterly: TSMC capex guidance, the advanced-node revenue mix, and CoWoS capacity commentary. A sustained foundry capex trajectory is the twelve-to-eighteen-month forward indicator for everything downstream.

Stage 2 — Memory and equipment capex (6-12 month lead)

High-bandwidth memory is the second structural bottleneck, and it has become a capex story in its own right. AI accelerators are memory-bound; if HBM bit-supply lags, GPU shipments lag regardless of how much logic capacity TSMC brings online. The signal that this constraint is binding now shows up directly in hyperscaler economics — memory is projected to consume roughly 30% of hyperscaler data center spending in 2026, a fourfold increase over the ~8% of 2023 (SemiAnalysis), and rising memory prices were explicitly cited by both Microsoft and Meta as a driver of higher 2026 capex guidance (Q1 2026 earnings). Microsoft attributed roughly \$25 billion of its 2026 capex to component and memory price inflation alone.

The equipment layer — Tokyo Electron, ASML, Applied Materials — provides the read on whether capacity additions are actually being tooled. Equipment order books and shipment data lead fab output by six to twelve months. The metrics to track: SK Hynix and Samsung HBM capacity commitments, HBM bit-supply growth against demand, and lithography/deposition equipment bookings. This is the layer where my coverage of the Asia-Pacific semiconductor complex provides direct visibility into the constraint before it propagates west.

Stage 3 — Accelerator unit volume (3-9 month lead)

The accelerator count drives the rack count drives the power demand. Nvidia, AMD, and the custom-silicon programs — Google's TPU, Amazon's Trainium, Meta's MTIA, all routed through Broadcom and Marvell — convert wafer and memory supply into deployable compute. Nvidia's data center revenue and forward visibility, plus Broadcom's custom-ASIC backlog tied to hyperscaler programs, are the read on how much compute is actually shipping versus reserved. Because a meaningful share of TSMC's high-performance-computing volume is Broadcom pass-through serving Alphabet and Meta, the foundry mix and the accelerator-vendor order books should be read together. The metric to track: accelerator unit volume and ASPs, custom-silicon ramp rates, and networking attach (which has been growing faster than compute as cluster sizes scale).

Stage 4 — Hyperscaler capex commitment (1-6 month lead)

This is the layer the market treats as leading and that is in fact lagging. By the time the Big Four hyperscalers report capex on an earnings call, the upstream signal is six to twelve months old. The current print is nonetheless the corroboration the thesis needs: combined 2026 capex guidance approaching \$700 billion across Microsoft, Alphabet, Amazon, and Meta, up roughly 75% from 2025, with approximately 75% directed at AI infrastructure (Financial Times, Q1 2026). Amazon alone guided to \$200 billion; Alphabet to \$175-185 billion; Microsoft to roughly \$190 billion on a calendar-year basis; Meta to a raised \$125-145 billion. The capital intensity is historically unprecedented — 45-57% of revenue (CreditSights) — and is increasingly debt-funded, with roughly \$120 billion of new debt raised in 2025 (Bank of America) and Morgan Stanley and JPMorgan projecting up to \$1.5 trillion of AI- and data-center-related issuance over the next five years.

The signal within the signal is the forward commitment that does not appear on the capex line: Microsoft's commercial remaining performance obligations stood at roughly \$627 billion and Google Cloud's backlog above \$460 billion (Q1 2026), contracted future revenue that underwrites the buildout. The metric to track: combined hyperscaler capex run-rate, the AI-infrastructure share, and the backlog/RPO figures that indicate whether the spend is demand-backed.

Stage 5 — Data center demand and pre-leasing (real-time, where REIT economics live)

The demand finally arrives at the real estate layer as take-up of powered, cooled, AI-ready capacity — and it arrives into a supply base that is structurally short. Powered-shell vacancy in the primary US markets has compressed to the low single digits, with Northern Virginia below 5% and the tightest submarkets near 1%. Wholesale colocation for hyperscale and AI tenants accounts for the majority of new leasing volume, and the binding constraint has shifted decisively from capital or chips to power: utilities cannot deliver new substation capacity fast enough for AI training clusters requiring 100-plus MW per campus, and a large share of announced projects is slipping on power — roughly a third of 2025 capacity was delayed or deferred, with 30-50% of 2026 capacity at risk (Sightline Climate). Demand is spilling from saturated primary markets into Atlanta, Phoenix, and Dallas as a direct consequence.

The metrics to track here are the ones that drive REIT cash flow directly: pre-leasing rates and take-up in MW by market, powered-shell availability, releasing spreads on renewals, and the development pipeline's pre-leased percentage. Data center power demand is projected to reach roughly 1,000 TWh globally in 2026 — comparable to Japan's total electricity consumption (roughly double Germany's) — which frames the scale of the physical constraint.

The framework's payoff

The five stages do not move in lockstep; they move in sequence, and the sequence is the edge. Throughout 2024 through 2026, Stages 1 through 3 ran consistently ahead of Stage 4 consensus — foundry and memory capacity commitments implied more downstream compute than hyperscaler guidance had yet ratified, and the hyperscaler guidance then repeatedly revised upward to meet the upstream signal. For a real estate investor, the practical consequence is that the demand surprise at Stage 5 is partially forecastable twelve to eighteen months in advance by tracking Stages 1 and 2. When advanced-node and HBM capacity commitments accelerate, data center take-up accelerates with a lag, and the REITs with the most pre-funded, power-secured development pipeline capture it first.

The thesis does not rest on AI demand growing forever. It rests on a narrower and more defensible claim: that for the duration of the current cycle, supply — gated by power interconnection queues and advanced-node capacity that grows only ~69% through 2028 — cannot catch demand, and that this imbalance is the precondition under which data center real estate cash flows compound and cap rates hold. The downside is cushioned not by demand optimism but by the physical impossibility of building supply quickly.

The lead, demonstrated

The framework is not just a set of stages — it is a sequence with a measurable lag, and one explicit chain shows it. The figures below place the five stages on a common 2024–2026 timeline and trace a single anchor — TSMC's January 2025 capacity commitment — through to the REIT-level records of late 2025 and early 2026.

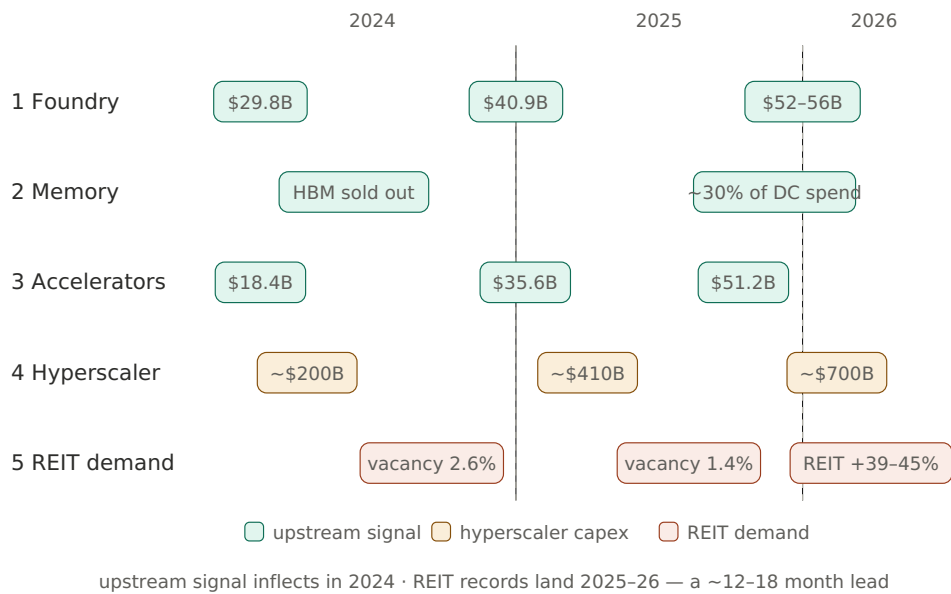


Exhibit 1. Upstream foundry, memory and accelerator signals inflect in 2024; hyperscaler capex steps up through 2025; data-center demand and REIT returns set records in 2025-26 — a ~12-18 month lead.

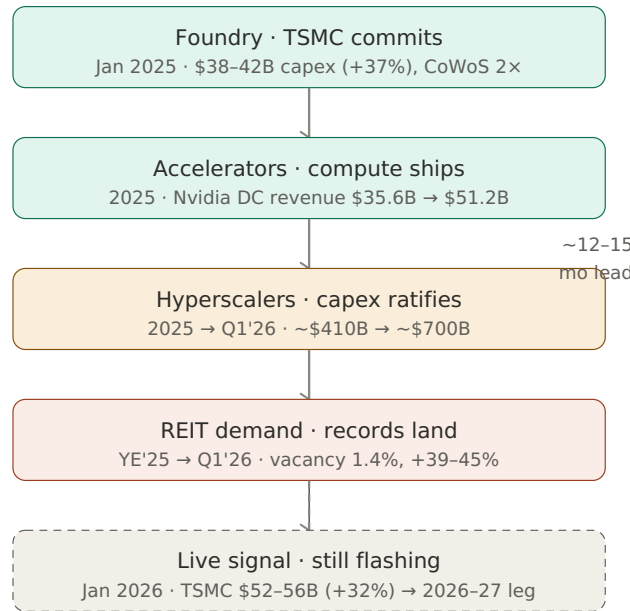


Exhibit 2. One worked example: TSMC's January 2025 commitment (\$38-42B capex, CoWoS doubling) → accelerator shipments → hyperscaler capex ratification → record REIT demand by Q1 2026, a ~12-15 month lead. TSMC's January 2026 commitment is the live forward signal pointing into 2026-27.

These five stages do not merely correlate — the upstream layer *predicts* the downstream one, and the January 2025 chain is the proof. TSMC's commitment that month was not a coincident readout of demand already visible; it was a forward, multi-year, largely irreversible capital decision, taken against customer prepayments and capacity reservations, for silicon that physically cannot become leased, powered floor for twelve to eighteen months. The lag is mechanical, not statistical: a wafer committed in Q1 2025 becomes a packaged accelerator by mid-to-late 2025, a hyperscaler purchase order by year-end, and occupied megawatts in 2026. The record colocation demand of late 2025 and Q1 2026 is, quite literally, the arrival of the capacity TSMC priced in January 2025 — which is why the foundry commitment is a leading indicator of REIT take-up rather than a confirmation of it.

The obvious objection is that these series simply rose together inside one AI boom, so the apparent lead is an artifact. That misreads the causality. Advanced-node and advanced-packaging capacity is the binding constraint on how much AI compute can exist; it gates everything downstream, so its commitment must precede — not accompany — the demand it enables. And the lead survives being public: the investors who price data-center REITs do not model foundry capex or CoWoS unit counts, and the translation from a packaging-capacity decision in Hsinchu to same-store NOI in Northern Virginia spans enough specialist domains that it is not arbitrated into REIT prices in real time. That is precisely what makes it tradable. By the same logic, TSMC's January 2026 commitment — another 32% step-up and a second doubling of CoWoS — is a forward read on a demand leg that will not fully reach the real estate layer until 2026-27.

The dated chronology behind both exhibits:

Stage	2024	2025	2026
1 · Foundry (TSMC capex)	\$29.8B	\$40.9B (guided Jan'25 at \$38-42B)	\$52-56B (guided Jan'26; CoWoS 2×)
2 · Memory / HBM	HBM sold out	HBM sold out	memory ~30% of hyperscaler DC spend
3 · Accelerators (Nvidia DC rev.)	\$18.4B (qtr rep. Feb'24)	\$35.6B (Feb'25); FY +142%	\$51.2B (Nov'25, "cloud GPUs sold out")
4 · Hyperscaler capex (combined)	~\$200B	~\$410B	~\$700B (~75% AI-directed)
5 · DC demand / REITs	vacancy 2.6% (YE)	vacancy 2.3% (H1)	vacancy 1.4% (YE'25); REIT +39-45% TTM

Sources: TSMC Q4'24 / Q4'25 earnings (capex, CoWoS); Nvidia 8-K filings (data-center revenue); Reuters / FT / 24-7 Wall St. (hyperscaler capex); CBRE / JLL 2024-2026 (vacancy, absorption, returns). Full sourcing in the appendix.

Section 2 — Bottom-Up Demand Model & Sector Mapping

The Stage-4 capex signal translated into megawatts, and the megawatts set against deliverable supply.

Calculator: a working model accompanies this section — DC_Demand_MW_Bridge.xlsx. Set the three highlighted assumptions (facility share of capex, \$/MW basis, colocation share) and it returns the implied demand. On grounded defaults it produces ~3,150 MW/yr of colocation-addressable demand against the record 2,498 MW the market absorbed in 2025 — a 1.26× starting point. The verdict below reads that result; the model lets you reset the assumptions and test it yourself.

The translation to quantify. Convert the Stage-4 capex signal into incremental data-center demand, then into REIT-addressable take-up:

- Combined hyperscaler 2026 capex ~\$700B, ~75% AI-directed (~\$525B), up from ~\$410B in 2025 (*Reuters / FT / 24-7 Wall St., Q1 2026*).
- A large share is IT hardware (chips/servers), not building + power; isolating the *real estate* portion (powered shell, fit-out) matters — facility/power is commonly ~25-40% of total capex (vs. silicon/servers), and that split drives the MW estimate.
- The bridge: facility \$ → MW (AI-ready cost commonly ~\$8-12M/MW) → square footage and annualized lease revenue at market rents → share addressable by third-party colocation vs. hyperscaler self-build.

The supply side (sourced): - North American colocation vacancy at a historic low ~2.3%, **<1% in the largest markets including Northern Virginia** (*JLL, 2025-26*); ~3/4 of the development pipeline pre-leased; crunch expected to persist through 2027. - **Power — not capital or chips — is the binding constraint:** average power-connection wait ~4 years, interconnection queues 3-5+ years, some utilities pausing new connections; ~1,000 TWh global DC electricity demand projected for 2026 (*IEA; JLL; Utility Dive, 2026*). - Demand spilling from saturated primary markets (NoVa, Santa Clara) into Atlanta, Phoenix, Dallas-Fort Worth and emerging low-cost-power states (Texas, Louisiana, Oklahoma, Idaho).

The absorption verdict. The bridge supports a clear conclusion: incremental demand structurally outruns deliverable supply, and the constraint is physical rather than financial. Even at the most conservative corner of the assumption grid — a 25% real-estate share of AI capex and only 40% addressable by third-party colocation — the 2026 commitment implies roughly 2,600 MW of incremental colocation demand, already at the record ~2,500 MW the market cleared in 2025; the grounded base case (1.26×) and the upper range (toward 2.9×) put demand well ahead of it. Supply cannot close that gap on the same clock: ~4-year grid-interconnection waits and 18-36-month campus build times mean the throttle is power delivery, not capital or chips, and the markets that win are the ones with secured power — the spillover into Texas, Phoenix, Atlanta and the low-cost-power states is demand routing around the constraint. The read is therefore confident in direction and conservative in magnitude: under every assumption tested, demand meets or exceeds the most absorption the market has ever delivered — the condition under which occupancy, rents and renewal spreads compound. The figure is assumption-dependent, which is why the calculator makes those assumptions explicit and auditable rather than burying them; the conclusion itself does not flip across the plausible range.

Section 3 — Public-Market REIT Analysis

The three public names on a common frame; sourced financials below (verify against each company's latest 10-K/supplemental).

Sector anchor: trailing-12-month total returns through ~April 20, 2026 — **DLR 39.0%, EQIX 41.0%, IRM 45.1%**, well ahead of the broader REIT sector (*JLL / Angel Investors Network, 2026*).

Comparison (sourced):

	DLR	EQIX	IRM
Model	hyperscale + interconnection	interconnection moat	storage (⅔ rev) + DC growth
FY2025 revenue	\$6.11B (+10%)	~\$9.2B (+7-8% cc)	\$6.8-6.9B (+12%)
2026 revenue (guide)	\$6.6-6.7B (+~10%)	\$10.12-10.22B (+10-11%)	\$7.83-7.93B (+14%)
2026 FFO / AFFO per share	Core FFO \$7.90-8.00 (+8%)	AFFO ~\$42-43 (+9-11%)	AFFO \$5.79-5.86 (+~14%)
Net debt / EBITDA	4.9× · BBB+/Baa2/BBB	~3-4× · investment-grade	4.9× net lease-adj. · sub-IG (~8× GAAP)
TTM total return → Apr '26	+39.0%	+41.0%	+45.1%
Dividend (approx. yield)	\$4.88 (~2.4%)	~\$18.7 (~2%)	~\$3.46 (~4%) · +10%/yr
The angle	hyperscale operating leverage	interconnection moat, switching costs	storage cash flow + DC re-rate

Sources: DLR / EQIX / IRM FY2025 results and 2026 guidance (8-Ks / supplementals, Q4'25-Q1'26); TTM total returns

through ~Apr 20 2026. *EQIX leverage and dividend yields are approximate.*

Digital Realty (DLR) — sourced: - FY2025 revenue \$6.11B (+10%), Adj EBITDA \$3.34B, Core FFO \$7.39/sh (+10%); 2026 guidance Core FFO \$7.90-8.00, revenue \$6.6-6.7B, Adj EBITDA \$3.6-3.7B (*DLR Q4'25, 2/5/26*). - ~300+ data centers across 50+ metros; hyperscale + 0-1MW/interconnection. Leverage ~4.9x net debt/EBITDA; BBB+/Baa2/BBB (stable). Price ~\$203.62 (4/20/26); market cap >\$60B; dividend \$4.88. - Valuation detail in the accompanying DLR three-statement model (Project 1).

Equinix (EQIX) — sourced: - 2026 guidance: revenue \$10.12-10.22B (+10-11%); AFFO \$4.20-4.28B (+9-11%) (*EQIX Q4'25; DCD, 2026*); record 2025 capacity (23,250 cabinets, >90MW xScale).

Iron Mountain (IRM) — sourced: - Best TTM return of the three (~45.1%); data-center growth on top of recurring records-storage cash flow; 2026 revenue guide ~\$7.6-7.8B, Adj EBITDA ~\$2.9B (data-center segment ~\$1B revenue) (*IRM Q4'25; DCD, 2026*).

The comparative verdict and ranking. On a risk-adjusted basis the order is **Equinix first, Digital Realty second, Iron Mountain third** — a quality-and-balance-sheet ranking, not a momentum one. Equinix takes the top slot because its interconnection fabric is the most durable moat in the group: tenants deployed across its metros and exchanges face real switching costs, churn is structurally low, and it carries the cleanest investment-grade balance sheet at the lowest leverage (~3-4x net debt/EBITDA), which gives it the most defensible cash flows in a sector being repriced on growth. The trade-off is the slowest per-share growth and the lowest yield — quality is not cheap. Digital Realty ranks second as the purest investment-grade expression of the hyperscale build-out: 300-plus facilities across 50-plus metros, direct hyperscale operating leverage, and a development pipeline geared straight to the capex cycle this thesis tracks, with its valuation worked in depth in the accompanying three-statement model. Iron Mountain ranks third not for weak performance — it posted the fastest revenue growth (+14%) and the best trailing return (+45%) of the three — but because that return carries the most risk: it is sub-investment-grade, materially more levered on a GAAP basis (~8x), and roughly two-thirds of revenue is still legacy records storage, which makes the data-center re-rate an optionality layered on a slower base rather than the core engine. The ranking is a risk-adjusted lens; an investor weighting growth and momentum over balance-sheet quality would reasonably move Iron Mountain toward the top.

An Asia-Pacific overlay fits the same framework and extends the basket into the region's tightest markets through Keppel DC REIT (SGX: AJBU) — a Singapore-, Japan- and Europe-weighted, hyperscale- and AI-focused data-center REIT. In FY2025 it grew gross revenue ~42% and net property income ~47% on roughly S\$1.1 billion of acquisitions, lifted DPU 9.8% to 10.381 cents (~4.6% yield) on 45% rental reversions, and carried conservative ~35% gearing — momentum that continued into Q1 2026 (DPU +13% year-on-year, ~96% occupancy). It adds Asian supply-tightness and a currency dimension to an otherwise US-concentrated thesis.

Section 4 — Risk Framework & Contrarian Scenarios

Three risks, argued both ways, each with quantified AFFO downside scenarios.

The risk assessment, both sides. Three risks bound the thesis, and each cuts in two directions. The most weighted is an AI-capex digestion — that 2027-28 demand undershoots the capacity now being committed. The bear case is real and quantifiable: capital intensity at 45-57% of revenue with free cash flow turning negative (Amazon's is guided negative in 2026), funded increasingly by debt (~\$120B raised in 2025 against up to ~\$1.5T of projected issuance), means a demand air-pocket would hit lease-up and pricing directly. A roughly -20% AFFO scenario follows from a one-to-two-year pause in net new leasing that stalls development yields and softens renewal spreads; a -40% scenario adds cap-rate expansion as the sector's growth multiple compresses toward a utility multiple. The rebuttal is the contracted backlog — Microsoft's ~\$627B in commercial RPO and Google Cloud's >\$460B — which underwrites near-term demand even if forward orders cool, making this a timing risk more than a thesis-breaker. **Net: medium.** The second risk is power binding harder than expected. Near-term it is real-estate-positive, because scarcity is exactly what supports rents and cap rates; the danger is a prolonged interconnection freeze that converts scarcity into a hard operating ceiling and invites rate-pass-through and political backlash. It is genuinely double-edged rather than one-directional — **medium**, and the one risk that is partly the thesis's own engine. The third, custom-silicon disintermediation — TPUs, Trainium and MTIA routed through Broadcom and Marvell compressing dollars-per-megawatt — is the lowest-weighted: cheaper compute per rack lowers spend per megawatt, but it does not reduce the megawatts of powered, cooled space those accelerators require, and floor space is what the REITs actually lease. **Low.** The downside scenarios are most sensitive to the capital-intensity and debt-funding lines; those are what turn a timing wobble into something structural, and they are the figures to monitor.

Section 5 — Thesis & Portfolio Positioning

The stake, the recommendation, and the signals that would change it.

The stake. Data-center REITs are the cleanest public-market exposure to the AI capital cycle without single-name silicon risk, and the binding constraint — power, not chips — is what cushions the downside that a pure technology-demand bet would carry.

The recommendation: own the basket. Hold the data-center REIT complex as a basket rather than a single name, sized as a 5-10% thematic AI-infrastructure sleeve, and built into the power-headline selloffs that periodically hit the group rather than chased after its ~40% trailing run. The entry discipline matters because the names are not cheap; the way in is to accumulate on weakness, not to pay up at the highs. Within the sleeve, weight toward quality on the risk-adjusted ranking from Section 3 — Equinix as the core holding, Digital Realty as the investment-grade, thesis-purest expression (its intrinsic valuation is developed in the accompanying three-statement model, on assumptions still being finalized), and Iron Mountain as a smaller, higher-beta position sized for its leverage rather than its trailing return.

Monitoring dashboard. The framework's payoff is a forward-looking watchlist that should move ahead of the REIT prints: TSMC capex, advanced-node mix and CoWoS commentary (the twelve-to-eighteen-month lead, Stage 1); HBM bit-supply and equipment bookings (Stage 2); accelerator volumes and networking attach (Stage 3); hyperscaler capex and the RPO/backlog that underwrites it (Stage 4); and at the real estate layer, pre-leasing and MW take-up, powered-shell availability and renewal spreads (Stage 5).

What would change the call. The thesis is falsified, not merely dented, in two cases: if foundry capex decelerates while REIT leasing stays hot — that would break the lead the whole thesis rests on — or if power constraints ease materially faster than demand, removing the scarcity that holds cap rates and rents. Either signal is a cut; their absence, on a pullback, is an add.

Appendix — Sourcing & Fact-Check Notes (audit, May 2026)

Figures in this draft audited against Q1 2026 primary/reputable sources.

Verified (clean): TSMC 2026 capex \$52-56B (~32% YoY), 70-80% to advanced nodes (7nm and below), CoWoS doubling (TSMC Q1'26). Combined hyperscaler 2026 capex ~\$700B (from ~\$410B 2025); Amazon \$200B / Alphabet \$175-185B / Microsoft ~\$190B; ~75% AI-directed; Microsoft \$25B component-inflation; Microsoft RPO \$627B; Google Cloud backlog >\$460B (Reuters / FT / 24-7 Wall St. / om.co). REIT TTM total returns DLR 39.0% / EQIX 41.0% / IRM 45.1% through ~Apr 20, 2026 (JLL / Angel Investors Network). NoVa colocation vacancy <1% in tightest submarkets; power the binding constraint; ~4-yr power-connection waits; ~3/4 of pipeline pre-leased; ~1,000 TWh global DC electricity demand 2026 (IEA; JLL; Utility Dive).

Corrections applied in this draft: (1) Meta's raised 2026 capex is **\$125-145B** (from \$115-135B), not \$115-145B. (2) ~1,000 TWh is **comparable to Japan's** electricity consumption, not Germany's (~500 TWh).

Sourced since the audit (figures pinned or updated): TSMC's 74% is its reported advanced-technology (7nm and below) share of wafer revenue — a different metric from its HPC platform revenue (~58%); both are correct (TSMC Q4'25). Advanced-node capacity +69% through 2028 (SEMI 300mm Fab Outlook, 2Q25). Memory ~30% of hyperscaler capex in 2026, ~4× the ~8% of 2023 (SemiAnalysis). Capital intensity 45-57% of revenue (CreditSights). Hyperscaler new debt updated to ~\$120B raised in 2025 (Bank of America), with up to ~\$1.5T of AI/data-center issuance projected over five years (Morgan Stanley; JPMorgan). The power-delay figure was softened from "~40%" to the sourced range: ~26% of 2025 capacity slipped and ~10% was deferred, with 30-50% of 2026 capacity at risk (Sightline Climate).

Independent analysis prepared for portfolio purposes. Views are my own and do not represent those of any employer. Not investment advice.